

The Intersection of Place and Game Design: The Case of Two Classic African Board Games

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Introduction

The concept of game design is often discussed relative to video games or digital games. “Game design” refers to the design of the gameplay experience. It includes how the game should work, what players will do, game characters, players’ interaction with these characters, and how the game will look.¹ For Mary Flanagan, game design is a creative process, a work of art, with aesthetics and values, that occurs in places or settings and engages the designer in a reflection on “humanistic themes or values” in the design process.² Although it can be argued that when it comes to board games, some developers have adopted video game or digital game design processes,³ researchers and designers such as Elizabeth LaPensée and Outi Laiti have suggested design processes for indigenous games.⁴ These approaches to game design focus on centering indigenous people, knowledge, and ways of doing—thus, enhancing indigenous people’s self-determination. Talking board game design, LaPensée explicates that it should involve collaboration with and contribution from indigenous community members.⁵ Laiti proposes the indigenous game design model to center indigenous people and knowledge.⁶ Indigenous communities design and use games as spaces to share their ways of knowing, “as learning and reinforcing community teachings is built into the game’s mechanics.”⁷

Similarly, Steve Craig, examining ancient games, including African games, writes that games developed in various cultures around the world were also designed as “a means of learning and mastering vital survival skills.”⁸ This article builds on this research on ancient games and indigenous game design to examine the design of historical or early African board games—also referred to as classic African board games—such as *Songo* or *Owela*. The historic and ancient nature of these games may make discussions about design in relation to them difficult. Still, as Flanagan explains, “early board games remain important to . . . [contemporary] designers not only because they reflect cultural notions of a given time but also because their play patterns continue to influence contemporary design.”⁹ These games have been in existence for thousands of years and have been discussed by anthropologists and

- 1 Robert Zubek, *Elements of Game Design* (Westchester, MA: MIT Press, 2020).
- 2 Mary Flanagan, *Critical Play: Radical Game Design* (Cambridge, MA: MIT Press, 2009).
- 3 Haider Akmal and Paul Coulton, “Research through Board Game Design,” in *Proceedings of the 4th Biennial Research through Design Conference*, 2019, <https://eprints.lancs.ac.uk/id/eprint/131465>.
- 4 Elizabeth LaPensée, “Self-Determination in Indigenous Games,” in *Routledge Companion to Media Studies and Digital Humanities* (Abingdon: Routledge, 2017), 128–37; Outi Laiti, “Old Ways of Knowing, New Ways of Playing—The Potential of Collaborative Game Design to Empower Indigenous Sámi,” PhD diss., University of Lapland, 2021, <https://urn.fi/URN:ISBN:978-952-337-249-8>.
- 5 Elizabeth LaPensée, “Indigenous Board Game Design in the Gift of Food,” *Analog Game Studies* 11, no. 1 (2016), <http://analoggamestudies.org/2016/03/indigenous-board-game-design-in-the-gift-of-food/>.
- 6 Laiti, “Old Ways of Knowing.”
- 7 Ibid., 5.
- 8 Steve Craig, *Sports and Games of the Ancients* (Westport, CT: Greenwood, 2002), 296.
- 9 Flanagan, *Critical Play*, 63.

archaeologists.¹⁰ Exploring design relative to African classic board games is relevant given the learning opportunities provided by board games, understanding that game design reflects cultural notions of a specific time, opportunities created for culturally appropriate game-based learning, and indigenous game design research.¹¹

The complexity of the gameplay indicates that these games involved a well-thought-out design process.¹² However, this article does not intend or attempt to uncover the process used by game designers of classic African games. Rather, we delve into the interaction of design and place to make sense of the design of these classic board games. Simultaneously, I draw on the players' interpretations of these games to connect and interact with their histories and memories. Design involves interaction with values, ways of doing, ways of knowing, ways of thinking, and material objects available in the location where the design occurs. Using two classic African board games, *Songo* from Cameroon, central/west of Africa, and *Owela* from Namibia, south of Africa, we explore and discuss design in relation to place and through the lenses of the game players.

In the literature, these games are often identified as Mancala, which is a generic term referring to classic African board games with rows and holes, played with seeds/stones distributed in a clockwise or counterclockwise direction.¹³ However, we argue that lumping these classic board games with the generic term "Mancala" takes away the complexities of the design and peculiar rules of these games.¹⁴ Hence, the generic term "Mancala" is not used in this article, but the name given by the communities that designed each game.

These games lend themselves well in our endeavor to understand game design of classic African board games and the role of place in shaping design, particularly game design. In the next sections, we introduce background literature for exploring the connection between place and design, then introduce the classic African board games *Songo* and *Owela*. We build on architectural and urban design literature and use the literature as a framework to show how place and design interact in these games. Using the terms "board games," "design," and even "board game designers" to discuss these games is challenging. As Kwasi Wiredu points out, using European concepts or Western concepts to make sense of African worlds, in this case games, is challenging.¹⁵ For instance, board games are often defined as games with a set of rules and played on a board with several pieces that allow physical interactions with other players.¹⁶ Moreover, boards are flat pieces of wood or other hard material, which is not the case for the games studied here. Based on this understanding of board, these African games would likely not fit the category of a board game as does *Chess* or *Warhammer 40,000*. It therefore becomes difficult to classify these games as board games. Yet for Eliso Macamo, the

- 10 Alex de Voogt, "Strategic Games in Society: The Geography of Adult Play," *International Journal of Play* 6, no. 3 (2017): 308–18, <https://doi.org/10.1080/21594937.2017.1382986>; Paul Gerdes, "Mathematics in Africa South of the Sahara," in *Encyclopaedia of the History of Science, Technology, and Medicine in Non-Western Cultures*, ed. Helaine Selin (Dordrecht: Springer, 2008), 284–85.
- 11 Nancy B. Sardone and Roberta Devlin-Scherer, "Let the (Board) Games Begin: Creative Ways to Enhance Teaching and Learning," *Clearing House: A Journal of Educational Strategies, Issues and Ideas* 89, no. 6 (2016): 215–22, <https://doi.org/10.1080/00098655.2016.1214473>; Flanagan, *Critical Play*; LaPensée, "Indigenous Board Game Design."
- 12 Rebecca Y. Bayeck, "Exploring the African Songo Game and How Gameplay Enhances Multiple Literacies among Adult Players in Cameroon and the United States," PhD diss., Pennsylvania State University, 2019, <https://etda.libraries.psu.edu/catalog/16327ryb105>; Alex de Voogt, *A Question of Excellence: A Century of African Masters* (Trenton, NJ: Africa World Press, 2005).
- 13 Voogt, *A Question of Excellence*.
- 14 Rebecca Y. Bayeck, "Understanding Computational Thinking in the Gameplay of the African Songo Board Game," *British Journal of Educational Technology* 55, no. 1 (2024): 259–76, <https://doi.org/10.1111/bjet.13353>
- 15 Kwasi Wiredu, "Can Philosophy Be Intercultural? An African Viewpoint," *Diogenes* 46, no. 184 (1998): 147–67.
- 16 Rebecca Y. Bayeck, "Examining Board Gameplay and Learning: A Multidisciplinary Review of Recent Research," *Simulation & Gaming* 51, no. 4 (2020): 411–31, <https://doi.org/10.1177/1046878119901286>.

issue is not necessarily in the use of Western or English concepts but in the ideas or assumptions underlying these concepts.¹⁷ Discussing African games using Western and English terms implies understanding and, in this article, defining the ideas that make up board games and design. Consequently, conceiving board games as games with playing pieces and surfaces that facilitate interactions among players explains their classification here as board games.¹⁸

Game Design and Culture

Game design researchers contend that values, such as cultural values or social values, are embedded in games.¹⁹ Design is not an incident or a coincidence, but a well-thought-out and planned process that consists of different steps or phases. For instance, the iterative design process consists of steps such as (a) setting a design goal; (b) developing the basic rules and assets necessary for the goal (e.g., tokens, characters); (c) playtesting the game; (d) revising based on feedback from the playtest; and (e) repeating the previous steps to ensure that the game is playable and captivating. Colleen Macklin and John Sharp revisit this design process as follows: (a) conceptualization, which is the process of generating the idea, prototyping, or the step where the idea takes shape; (b) playtesting or testing the game to see how it functions; and (c) evaluating based on the results from playtesting.²⁰ Integrated into this design process are design values, or the values and components a game designer wants to embody in the game and its play.²¹ Yet Flanagan's critical play method explicitly makes human concerns, values, or concepts a fundamental part of the design process.²² The critical approach to game design consists of (a) setting a design goal/mission statement and values goals; (b) developing rules and constraints that support values; (c) designing for many different play styles; (d) playtesting with diverse audiences; (e) verifying values and revising goals; and (f) repeating the process to ensure that the game supports the values set and is playable. Although culture is not clearly mentioned in this critical approach, Mary Flanagan and Helen Nissenbaum add that design values can reflect personal, political, cultural, social, and ethical values.²³ Indeed, games are

legitimate forms of . . . human expression, [with] cultural importance, and the ways games reflect the norms and beliefs of their surrounding cultures is essential to understanding both games themselves and the insights they may provide into human experience . . . board games [are] one of several artifacts of material culture used to trace social practices and beliefs.²⁴

Therefore, culture is nested in the game design process. Culture is as Clifford Geertz defines it: "an historically transmitted pattern of meanings embodied in symbols, a system of inherited conceptions expressed in symbolic forms by means of which men

17 Elisio Macamo, "Translation as Theory from Within," Pennsylvania State University Global Virtual Forum, November 10, 2023, <https://gvf.la.psu.edu>.

18 Ibid., 18.

19 Mary Flanagan and Helen Nissenbaum, *Values at Play in Digital Games* (Cambridge, MA: MIT Press, 2014).

20 Colleen Macklin and John Sharp, *Games, Design and Play: A Detailed Approach to Iterative Game Design* (Brooklyn, NY: Addison-Wesley Professional, 2016).

21 Ibid.

22 Flanagan, *Critical Play*.

23 Flanagan and Nissenbaum, *Values at Play in Digital Games*.

24 Flanagan, *Critical Play*, 67.

communicate, perpetuate, and develop their knowledge about and their attitudes toward life.”²⁵ Games can be seen as symbols that embody the knowledge and values of a community. Game design research reveals that board games have accurately mirrored the cultural practices of the locations in which they have been played.²⁶ Similar claims cannot be made about place regarding board game design. LaPensée argued for a culturally responsive design in her work on the design of indigenous board games.²⁷ For instance, having indigenous people as collaborators in the design process is important for the game to be an accurate representation of the community culture and to align with its values. Greg Loring-Albright studied the design of contemporary board games such as the Settlers of Catan when played in the United States, and how they connect players to the place of the historical settlement and simultaneous genocide of Native American tribes.²⁸ Yet the game keeps these communities invisible, which evidences the noncollaborative approach adopted by the game designers. A community-driven or collaborative development process where communities are also involved in all the steps (design, art, and writing) of board game design is critical.

In the context of classic African board games, adopting the game design concept to understand the making of these games is challenging. Indeed, Africans have used games such as Songo and Owela to express their philosophies, values, and approach to life and living of the ethnic groups that play the game.²⁹ Cega states that in the past, Songo underpinned a religious and mystical way of living life and communicated a particular existential philosophy to players and society.³⁰ The term “design” may seem misplaced and unrelated to the making of these games. Yet as previously noted, “design” is an English and/or Western concept; applying it to classic African board games to make sense of their creation has meaning given the authors’ conception of design. Scholars such as LaPensée and Laiti have addressed game design for games created by indigenous people.³¹ Acknowledging that indigenous people have a history of game-making, Laiti proposes the indigenous game design model.³² This model sees design as self-expression and centers on indigenous people’s culture and practices, such as storytelling.³³ Thus, the concept of design is not foreign to non-Western or non-European games. Game design is about self-expression; it’s also about the process of making a game.³⁴

Furthermore, researching design among the Yoruba people of Nigeria, Olúwólé Òkékándé explains that works of arts, including a game such as *Ayo Olopon*, which consists of a board with twelve holes and a total of forty-eight seeds, follow the Yoruba mythology created by the “Ifa myth of earth creation design numerology.”³⁵ It is safe to talk about design relative to African classic games and argue with Òkékándé that the design process of these games is influenced by cultural norms, values, and practices rooted in

25 Clifford Geertz, *The Interpretation of Cultures* (New York: Basic Books, 1973), 89.

26 Flanagan, *Critical Play*.

27 LaPensée, “Indigenous Board Game Design.”

28 Greg Loring-Albright, “The First Nations of Catan: Practices in Critical Modification,” *Analogue Game Studies* 6, no. 1 (2015), <http://analoggames-studies.org/byline/greg-loring-albright/page/3/>.

29 Bayeck, “Exploring the African Songo Game”; Jean-Marcel Meka Obam, *Le jeu du Songo: Reflet du social* (Paris: L’Harmattan, 2008).

30 Cega, “Culture: Le jeu de Songoh,” *Aulech: Apprendre, Enrichir, et Partager sa Culture*, February 22, 2015, <https://www.aulech.com/songoh/>.

31 LaPensée, “Self-Determination in Indigenous Games,” 128–37; Laiti, “Old Ways of Knowing.”

32 Laiti, “Old Ways of Knowing.”

33 LaPensée, “Self-Determination in Indigenous Games,” 128–37; Laiti, “Old Ways of Knowing.”

34 Elizabeth LaPensée, Outi Laiti, and Maize Longboat, “Towards Sovereign Games,” *Games and Culture* 17, no. 3 (2022): 328–43, <https://doi.org/10.1177/15554120211029195>; LaPensée, “Indigenous Board Game Design,” 11.

35 Olúwólé T. Òkékándé, “A Semiotic Investigation of the Relations between Ifá and Yorùbá Indigenous Markets, Markets Location and Marketing Theories,” *Nokoko* 8 (2020): 24, <https://ojs.library.carleton.ca/index.php/nokoko/article/view/2791>.

local mythologies³⁶ Culture is an integral part of design in the African context.

Place and Game Design

The term “place” in game design often refers to the setting in the game, not to the place, environment, context, or setting where the game was designed. For instance, allusions to *Call of Duty: Modern Warfare* setting, context, and even place in the literature point to World War II; *Grand Theft Auto: San Andreas* is Los Angeles; and *Resident Evil 5* a West African city called Kijuju.³⁷ In the literature, attention is mostly given to the place or context re-created in the game, the virtual place or context, and not to the context or place where the game design happened. Place, context, and setting where games are designed are rarely discussed in the literature. Overlooking place in game design may leave out non-digital activities and other ways of thinking, doing, or knowing that may contribute to the design of non-digital games. However, Benjamin Stokes argues that these games are inserted into the social fabric of place.³⁸ The design of these games is influenced by local places, and the success of *Pokémon Go* is evidence of the influence and importance of place in game design.

This article discusses the concept of design and conceives it as the process of creation that occurs in a certain time, place, situation, and culture; it takes into account or aligns with the realities, norms, and technology/material available.³⁹ We also dive into the literature on place to define and explain the interactions between place and design.⁴⁰ Furthermore, we build on game design literature that already describes games as systems of information, cultural artifacts, and expressions of cultural ways, habits, or customs, deemed to represent cultural norms and practices.⁴¹ As systems of information, games can certainly give information about places, in this case, places of creation, and can represent ways or customs of a place. We contend that there is no design process completed in a vacuum. Game design involves an interaction with values, cultural practices, and objects in the place where the process of design occurs. Therefore, place participates in the creation of unique designs that make the designed object identify with the place and be the identity of the place. A designed object may reflect the material aspect and practices of a place or region, given that place and/or cultural context can participate in the design process.

Drawing on the game design literature in the following paragraphs, we define place and discuss the interplay of place and design in two classic African board games, *Songo* and *Owale*. We focus on the aesthetics of these games; how they look or present themselves to the players, which includes the game pieces.

36 Ibid.

37 Holger Pötzsch and Vít Šisler, “Playing Cultural Memory: Framing History in *Call of Duty: Black Ops* and Czechoslovakia 38–89: Assassination,” *Games and Culture* 14, no. 1(2019): 3–25, <https://doi.org/10.1177/1555412016638603>; André Brock, ““When Keeping It Real Goes Wrong”: *Resident Evil 5*, Racial Representation, and Gamers,” *Games and Culture* 6, no. 5 (2011): 429–52, <https://doi.org/10.1177/1555412011402676>.

38 Benjamin Stokes, *Locally Played: Real-World Games for Stronger Places and Communities* (Cambridge, MA: MIT Press, 2020).

39 Flanagan, *Critical Play*.

40 Hernan Casakin and Fátima Bernardo, *The Role of Place Identity in the Perception, Understanding, and Design of Built Environments* (Oak Park, IL: Bentham Science, 2012); Nicole Lotz and Derek Jones, “Social Engagement in Online Design Pedagogies,” in *Proceedings of the 3rd International Conference for Design Education Researchers*, ed. Erik Bohemia, Ingvild Digranes, and Robin Vande Zande (Chicago, 2015), 1645–68.

41 Flanagan, *Critical Play*; Macklin and Sharp, *Games, Design and Play*.

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- 42 Bayeck, "Exploring the African Songo Game"; Setha Low, *Spatializing Culture: The Ethnography of Space and Place* (New York: Taylor & Francis, 2017).
- 43 Low, *Spatializing Culture*.
- 44 Robert A. Ellis and Peter Goodyear, "Models of Learning Space: Integrating Research on Space, Place and Learning in Higher Education," *Review of Education* 4, no. 2 (2016): 149–91; Arijit Sen and Lisa Silverman, eds., *Making Place: Space and Embodiment in the City* (Bloomington: Indiana University Press, 2014).
- 45 Jonathan D. Sime, "Creating Places or Designing Spaces?," *Journal of Environmental Psychology* 6, no. 1 (1986): 50.
- 46 Joseph Pierce, "How Can We Share Space? Ontologies of Spatial Pluralism in Lefebvre, Butler, and Massey," *Space and Culture* 25, no. 1 (2022): 20–32, <https://doi.org/10.1177/1206331219863314>; Joseph Pierce and Deborah G. Martin, "Placing Lefebvre," *Antipode* 47, no. 5 (2015): 1279–99.
- 47 Massey Doreen, *For Space* (London: Sage, 2005).
- 48 Henry Lefebvre, *The Production of Space* (Cambridge MA: Blackwell, Loring-Albright, 1991), 101.
- 49 Pierce and Martin, "Placing Lefebvre."
- 50 Bayeck, "Exploring the African Songo Game"; Ron Thompson et al., "Space, Place and Social Exclusion: An Ethnographic Study of Young People Outside Education and Employment," *Journal of Youth Studies* 17, no. 1 (2014): 63–78, <https://doi.org/10.1080/13676261.2013.793793>.
- 51 Setha Low, "Towards an Anthropological Theory of Space and Place," *Semiotica* 175 (2009): 22, <https://doi.org/10.1515/semi.2009.041>.
- 52 Simangele Dlamini, Solomon G. Tesfamichael, and Tholang Mokhele, "A Review of Place Identity Studies in Post-Apartheid South Africa," *South African Journal of Psychology* 51, no. 1 (2021): 121–33, <https://doi.org/10.1177/0081246320954040>.
- 53 Paul O. Agboola, Muhammad Farid Azizul, Mohd Hisyam Rasidi, and Ismail Said, "The Cultural Sustainability of Traditional Market Place in Africa: A New Research Agenda," *Journal of Rural Studies* 62 (2018): 87–106,

Defining Place

Definitions of "place" are multiple, and this concept is often opposed or assimilated to space. Place and space are defined differently across disciplines and researchers. Some scholars argue that place and space refer to the same idea, whereas others distinguish between the terms and see them intersecting.⁴² For Setha Low, the distinction (or lack of distinction) is often linked to the theoretical framing of the research problem.⁴³ Yet to situate place in the context of design, and in this study, we define "place" as a physical setting that is experienced by individuals.⁴⁴ Place "implies a strong emotional tie, temporary or lasting, between a person and a particular physical location."⁴⁵ Hence, embedded in our conception of place is the idea of physical location experienced by people, where physical location refers to the material environment, which may include mountains, rocks, cities, people, and cultural practices.⁴⁶ Connecting place with people's experience of a setting or physical location aligns with the understanding of space adopted by scholars such as Massey.⁴⁷ For Massey, space is dynamic and constructed through social relations and interactions of everything in it, such as "living beings, things, objects, works, signs and symbols."⁴⁸ Pierce and Martin also add interaction with the rocks, mountains, cities, and trees that exist.⁴⁹

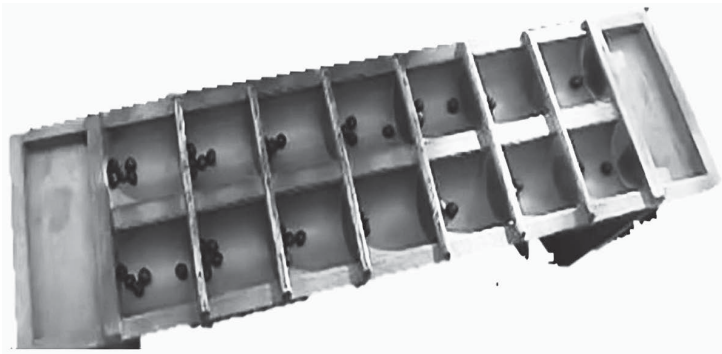
Drawing on these understandings of place and space, no distinction is made here between place and space. Place and space result from the interaction of the physical environment and social relations; they are both experienced and linked to identity because they give a sense of belonging, and cultural peculiarity as people provide meaning or ascribe feelings to a place/space.⁵⁰ As Low summarized it, place and space are embodied, and their materiality can be "metaphoric and discursive as well as physically located."⁵¹ Exploring the meaning of place in South Africa, Simangele Dlamini et al. connect place with identity and physical location.⁵² In a similar vein, Paul Agboola et al. in their work on the market in Nigeria link place to a location that manifests cultural values, enables social interaction, and facilitates participation in community activities.⁵³ In this study, place is a location with which a group or an individual identifies and interacts with memories, history, and objects of tangible and intangible heritage. Place can include a country, a region, a geographical location, or the "where-ness" of an object, or of a board game and the people, things, symbols, history, or environment where it interacts.

Methods

In this case study, we conceive board games as text and use textual analysis to understand the interaction between place and game design.⁵⁴ Textual analysis is an in-depth study of a text, which consists of using the text as a case study to inductively make sense of the text, which in this article is board games.⁵⁵ As Clara

Figure 1

Songo board game. Source: Photo by authors.



<https://doi.org/10.1016/j.jrurstud.2018.07.001>; Paul O. Agboola, Mohd Hisyam Rasidi, and Ismail Said, "The Influence of Open Space Utilization on Residents' Attachment with Community: A Case Study of Rural Market Square in Southwest Nigeria," *International Journal of Architectural Research: Archnet-IJAR* 11, no. 1 (2017): 44–66, <https://www.archnet.org/publications/12026>.

54 Clara Fernández-Vara, *Introduction to Game Analysis* (London: Routledge, 2015).

55 Ibid.

56 Ibid.

57 Logan Molyneux, "What Journalists Retweet: Opinion, Humor, and Brand Development on Twitter," *Journalism* 16, no. 7 (2015): 920–35, <https://doi.org/10.1177/1464884914550135>; Nikki Usher, Jesse Holcomb, and Justin Littman, "Twitter Makes It Worse: Political Journalists, Gendered Echo Chambers, and the Amplification of Gender Bias," *International Journal of Press/Politics* 23, no. 3 (2018): 324–44, <https://doi.org/10.1177/1940161218781254>.

58 Elfriede Fürsich, "In Defense of Textual Analysis: Restoring a Challenged Method for Journalism and Media Studies," *Journalism Studies* 10, no. 2 (2009): 238–52.

59 Ibid., 245.

60 Paul Booth, *Game Play: Paratextuality in Contemporary Board Games* (New York: Bloomsbury Publishing, 2015); Maria L. Leavenworth, "The Paratext of Fan Fiction," *Narrative* 23, no. 1 (2015): 40–60, <https://www.jstor.org/stable/24615501>.

61 Serges Mbarga Owona, *Le Jeu de Songo* [The game of Songo] (Paris: L'Harmattan, 2004); Jean-Marcel Meka Obam, *Le jeu du Songo: Reflet du social* [The game of Songo: Reflection of the social] (Paris: L'Harmattan, 2008).

62 Bayeck, "Exploring the African Songo Game"; Bonaventure Ondo Mve, *L'Owani et le Songa: deux jeux de calculs africains* [Owani and Songa: Two African games of calculation] (Saint-Exupéry: Centre culturel français, 1990).

Fernández-Vara explains, games can be approached as texts and can be analyzed as cultural creations that can be translated because they have meaning.⁵⁶ Textual analysis allows researchers to question the underlying assumptions of a text and go beyond the apparent content of the media or chosen text.⁵⁷ Elfriede Fürsich states, "textual analysis allows the researcher to discern latent meaning, but also implicit patterns, assumptions, and omissions of a text."⁵⁸ Text is understood in its broader, poststructural, sense as any cultural practice or object that can be "read."⁵⁹ Therefore, we used textual analysis to identify implicit patterns in the design of the classic African board games. Furthermore, we combined the textual analysis with semi-structured interviews of two game players. The interviews were treated as paratext, which refers to interpretation or meaning making beyond the text.⁶⁰ These interviews informed my analysis of the intersection of place and design in the classic African board games.

Place in Songo and Owale Board Game Design

The interplay of place and design is not new. As previously mentioned, research in fields such as architectural design has shown the interaction between place and design in their architectural practices. We explore this connection within the context of two board games.

Songo

Songo is a strategic board game played in Cameroon and other African countries such as Gabon and Equatorial Guinea.⁶¹ *Songo* (see Figure 1) consists of a rectangular wooden board, with two rows, fourteen holes, and seventy playing pieces, locally called *Ibang*, that is, seeds.⁶² *Songo* is played in a clockwise direction, and the winner is the player who captures the most seeds.⁶³

The game named *Songo* is played among the Ekang/Fang ethnic groups across Cameroon, Gabon, and Equatorial Guinea, with game rules reflecting the inherent values, way of life, and knowing of the Ekang/Fang ethnic group.⁶⁴ The *Songo* described in this article is discussed within the context of Cameroon.

Figure 2

Owela board game on the ground. Source: Gold Feather.



Owela

Owela is a strategic board game of the Ovakwanyama, Ovambandja, and Ovambalantu ethnic groups in Namibia. These groups belong to the Aawanbo ethnic group, located mostly in the northern part of Namibia. *Owela* consists of four rows, with thirty-two holes or more, with playing seeds from the marula tree,⁶⁵ referred to as *Eengobe* (cattle). *Owela* (see Figure 2) is played in a counterclockwise direction, and victory is based on the player or team that captures the most cattle.

Owela is a strategic game, which mirrors the way of life, thinking, and doing of the Ovakwanyama, Ovambandja, and Ovambalantu. This historical game is played in some parts of Angola as well. Here we focus on *Owela* played among the Aawambo ethnic group in Namibia.

The Interaction of Place and Design in Board Games

Songo (see Figure 1) is built or designed with wood and played with fruits from a tree locally called Njansang, ézezañ, or ézañ. The fruits of Njansang are edible and used as spices in different local dishes. Indeed, Njansang is one of the nontimber forest products of the Central Africa forests.⁶⁶ The tree has always been a strategic part of living for forest-dwelling people by providing subsistence needs and cash income.⁶⁷ As a result, it is integral to living for forest dwellers in Cameroon, as well as elsewhere in Central Africa.⁶⁸

- 63 Mbargo Owona, *Le Jeu de Songo* [The game of Songo]; Meka Obam, *Le jeu du Songo* [The game of Songo].
- 64 Meka Obam, *Le jeu du Songo*; Venant D. Zue Ntougou, *Le Songo: un jeu de stratégie du peuple Ekang* [Songo: a strategic game of the Ekang people] (France: Atramenta, 2021).
- 65 Anela Mafunzi, personal communication, March 20, 2022.
- 66 Louis Njie Ndumbe, Verina Ingram, Martin Tchamba, and Smith Nya, "From Trees to Money: The Contribution of Njansang (*Ricinodendron heudelotii*) Products to Value Chain Stakeholders' Financial Assets in the South West Region of Cameroon," *Forests, Trees and Livelihoods* 28, no. 1(2019): 52–67, <https://doi.org/10.1080/14728028.2018.1559107>.
- 67 Richard Eba'a Atyi et al., eds., *Étude de l'importance économique et sociale du secteur forestier et faunique au Cameroun: Rapport final* (Bogor, Indonesia: CIFOR, 2013); Julius Chupezi Tieguhong and Ousseynou Ndoye, "Transforming Subsistence Products to Propellers of Sustainable Rural Development: Non-Timber Forest Products (NTFPs) Production and Trade in Cameroon," in *Africa—Escaping the Primary Commodities Dilemma. African Development Perspective Yearbook*, ed. Karl Wohlmuth (Berlin: Lit, 2006), 107–37.
- 68 Ndumbe, Ingram, Tchamba, and Nya, "From trees to money: The contribution of Njansang (*Ricinodendron heudelotii*) products to value chain stakeholders"; Tieguhong, Chupezi, and Ndoye, "Transforming subsistence products to propellers of sustainable rural development: Non-timber forest products (NTFPs) production and trade in Cameroon."

The Ekang/Fang ethnic groups are in the south and center part of Cameroon, in the tropical lowland rainforest. In this environment, people are traditionally farmers; they depend on the forest for the wide range of products to meet their various needs.⁶⁹ The environment of the Ekang/Fang ethnic group explains the use of wood by these Ekang/Fang board game designers to create *Songo* (see Figure 1). The playing pieces also evidence the interaction of place and design.⁷⁰ Place, that is, the physical environment, with memories, tangible and intangible heritage, local practices, or way of life (e.g., eating/cooking) shaped the design of *Songo* board game design. According to the *Songo* player I interviewed, the game was designed and handed down to their generation by previous generations. Nicolas, speaking to the number of holes on *Songo* board, said: “quatorze cases c’est sa particularité, c’est comme ça qu’il est conçu, et chacun des deux adversaires a sept cases” [fourteen holes that is its peculiarity, that is how it’s been designed, and each player has seven holes]. Nicolas recognized one of the specificities in the design of *Songo* even though he could not explain why the game has a total of fourteen holes with seven holes for each player. Zue Ntougou discussed the origins of the game and explained that it represents an aspect of the Ekang ethnic group philosophy of life and living.⁷¹ The seven holes for each player may point to the Ekang people thinking about life and living in this world. Indeed, for the Ekang people who live off the land, their work—farming and fishing throughout different seasons of the year—is critical.⁷² An Ekang must strategically work to live comfortably, and this conception of life is embedded in *Songo*. Isañ, a *Songo* player, explains that the seven holes on each side of the game board represent the seven days of the week given to every human by the creator (i.e., Ntondobe) to strategically plan and/or work to reap the fruits of their labors. The five seeds point to the five fingers, which enable everyone to complete their daily work to be successful. The design of *Songo* appears to be situated in the philosophy of life and living of the Ekang people. Drawing on our conception of place as including a people’s way of life, in this case philosophy of life and living, place informed the design process as shown in the architectural and urban design literature.⁷³ The understanding of the local environment and practices sheds light on situatedness and place-informed board game design. The designers use wood (the Ekang live in the forest); the playing pieces (*Ibang*) are edible fruits of a tree locally used for different purposes; and the number of holes is rooted in the Ekang approach to live and living.

The design of *Owela* (see Figure 2) by the Awambo board game designers was also informed by the place or where-ness of the game creation. Anse explains that “in olden days holes of four rows were dug in the ground. Now, crafted in stones or wood.” The Awambo people rely mostly on a combination of agriculture and animal husbandry. In fact, wealth is measured by the

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- 69 Ebenezer K. Asaah, Zacharie Tchoundjeu, Roger R. B. Leakey, Bertin Takousting, James Njong, and Innocent Edang, “Trees, Agroforestry and Multifunctional Agriculture in Cameroon,” *International Journal of Agricultural Sustainability* 9, no. 1 (2012): 110–19, <https://doi.org/10.3763/ijas.2010.0553>.
- 70 Meka Obam, *Le jeu du Songo* [The game of Songo].
- 71 Ntougou, *Le Songo: un jeu de stratégie du peuple Ekang* [Songo: a strategic game of the Ekang people].
- 72 Brigitte Nga Ondigui, “Réécriture de soi chez les Ekang: entre savoir mémoriel et valeur identitaire du m’mouat (costume),” *Revista de Antropologia da UFSCar* 13, no. 2 (2021): 62–79.
- 73 Marcin Butlewski, A. Misztal, and N. Belu, “An Analysis of the Benefits of Ethnography Design Methods for Product Modelling,” *IOP Conference Series: Materials Science and Engineering* 145, no. 4 (2016): 1–9, <https://doi.org/10.1088/1757-899X/145/4/042023>; Rachel Jones, “Experience Models: Where Ethnography and Design Meet,” *Ethnographic Praxis in Industry Conference Proceedings*, no. 1 (2006): 82–93.



Figure 3
Owela board game, carved in stones.
Source: Manni Goldbeck.

number of cattle, sheep, or goats. The Aawambo live in a semiarid environment, consisting of extremely flat plains covered by white sands.⁷⁴ In the design process, the Aawambo used material available in their environment and thus dug holes in the ground (see Figure 2). The flatness of the land was appropriate for such a design.

The playing pieces used for gameplay are fruits from locally edible trees such as marula, which is available in the area where the game was designed. Furthermore, marula fruits are known to produce alcoholic beverages in Namibia.⁷⁵ The interaction of place with the game design is noticed in the name of the playing pieces *Eengobe*, that is, cattle, a reference to the Aawambo other means of survival. The Aawambo do not live only in a semiarid environment, they are also cattle herders, with cattle being a sign of wealth. Hence, the use of *Eengobe*/cattle is an indication of the entanglement of design with the lives of people, places, and things.⁷⁶ This reflects the interaction of the Aawambo ways of living/cultural practices with game design. The game design—holes in the ground and playing seeds—is informed by the place (plains, white sands, and the marula tree) of the designed object. Aawambo board game designers' understanding of the place material and local practices shaped the design process; this allowed them to create board games that mirror the place of design.⁷⁷ It is important to note, as Anse stated, that now *Owela* holes, particularly in cities, are crafted in wood or stone, as seen in Figures 3 and 4.

Interestingly, the shift in the visual design of *Owela* is still informed by the setting of the game. These holes are crafted into local materials, which confirms the situatedness of the design

74 Job S. Amupanda, "Pre-Colonial and Postcolonial Identity Formations in Namibia: An Exploration of the Origin of Nehale IyaMpingana's Dualism," *Journal of Namibian Studies: History Politics Culture* 23 (2018): 7–27, <https://namibian-studies.com/index.php/JNS/article/view/156/156>; Felician N. Hainduwa, Cori Ham, and Martina Meincken, "Impact of Fuelwood Quality and Quantity on the Energy Use of Rural Households in the Omusati Region, North-Central Namibia," *Southern Forests: A Journal of Forest Science* 78, no. 4 (2016): 299–306, <https://doi.org/10.2989/20702620.2016.1207134>.

75 Jane Misihairabgwi and Ahmad Cheikhoussef, "Traditional Fermented Foods and Beverages of Namibia," *Journal of Ethnic Foods* 4, no. 3 (2017): 145–53, <https://doi.org/10.1016/j.jef.2017.08.001>.

76 Isabelle Doucet and Hélène Frichot, "Resist, Reclaim, Speculate: Situated Perspectives on Architecture and the City," *Architectural Theory Review* 22, no. 1 (2018): 1–8, <https://doi.org/10.1080/13264826.2018.1418127>.



Figure 4
Owela board game carved in wood. Source:
Ndinelao Shikemini.

process and highlights the peculiarity of the design of these classic African board games. Unlike digital or video games, whose design elements may include raw materials imported from different locations, the design of these classic African board games is embedded in the where-ness of the built games. Where-ness includes the way of life, trees, local practices, and knowledge that inform the design of these games. The interaction of design and place in board games, particularly in these classic African board games, reveals analog technologies, and we dare say that technologies are not designed in a vacuum. As with any design or built-in-environment, these classic African board games draw and reflect the where-ness and tangible and intangible heritage of their designers, ethnic groups, and players. Interestingly, our analysis also reveals how players' interactions and perceptions of these games highlight the interplay of place and design. Players of these classic games also endeavor to connect their world, social connections, ancestries, histories, and memories through the material configurations of the game. While we argue that design is not a strange concept of the African approach to making items such as games,⁷⁸ we also notice how players interact with these games to connect to a place, that is, histories, memories, and intangible heritage specific to the ethnic group. At first glance, it may be difficult to see how the designers of *Owela* and *Songo* may inform the design of games today. By observing how players create a place as they use the game and its material configuration to interact with their histories/ancestries and practices, the relationship between place and design becomes evident.

Implication for Learning

As Flanagan and LaPensée show, there is something about making games that requires attention to the potential of the game to be

77 Galen Cranz, *Ethnography for Designers* (New York: Routledge, 2016); Heng Zhang, Rodney H. Matsuoka, and Yu-Jan Hunag, "How Do Community Planning Features Affect the Place Relationship of Residents? An Investigation of Place Attachment, Social Interaction, and Community Participation," *Sustainability* 10, no. 8 (2018): 2726, <https://doi.org/10.3390/su10082726>.

78 Òkékéwándé, "A Semiotic Investigation."

meaningful and enjoyable for players.⁷⁹ Building on these authors, we contend that paying attention to the design of board games has the potential to shape learning, that is, what is learned and how. The renaissance of board games in recent years has led to increased interest from researchers in exploring their teaching and learning potential.⁸⁰ This article indicates that researchers should also pay attention to the context, place, and even culture where the game is designed. Games are likely to reflect the cultural patterns of the place where they were developed, thus some games may be more relevant to some contexts and cultures. In the push for game-based learning, the question of culturally relevant game-based learning should also be considered. There is no question that game “design decisions influence situated learning outcomes for the broad diversity of learners within and beyond our classrooms.”⁸¹ Given the diversity of learners and the relationship between design and learning, it is important when using games for teaching and learning to also consider how the design of these games may affect learning outcomes. In other words, some games may be more relevant than others in some contexts or cultures.

Conclusion

In this article, we explored the interaction of game design and place with classic African board games. With game design being mainly discussed in video game and modern board game studies, this article attempts to expand research on game design to indigenous or historical games. The limited literature on board game design, particularly historical games, may be explained by their historic nature. Indeed, these classic African board games are some of the first games in the world, and the design process of these early designers can no longer be retrieved. Drawing on architectural design literature gives insights into the design process. Place is useful for understanding the design process of these classic African board games. Highlighting the intersection of place and game design adds to the literature on game design. This article reveals the need to draw on other fields to advance game design research; it helps situate classic African board games in the field of game studies and game design. Though the examination of *Songo* and *Owela* board game designs was limited to visual elements, it does point to the complex design process the Ekang/Fang and Aawambo game developers participated in, while allowing these games and designers to occupy territories in game design literature.

79 Flanagan, *Critical Play*; LaPensée, “Self-Determination in Indigenous Games.”

80 Bayeck, “Examining Board Gameplay and Learning,” 51; Sardone and Devlin-Scherer, “Let the (Board) Games Begin,” 89.

81 Douglas B. Clark, Emily E. Tanner-Smith, and Stephen S. Killingsworth, “Digital Games, Design, and Learning: A Systematic Review and Meta-Analysis,” *Review of Educational Research* 86, no. 1 (2016): 117, <https://doi.org/10.3102/0034654315582065>.